

SHRAOSHREE GHOSH

Software Engineer

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SUMMARY

Software Engineer with 4+ years building data-processing applications, internal tools, and backend services in Python, SQL, and C++, now completing an M.S. in Business Analytics. Supported 100+ business-critical workflows processing 10M+ telecom records (~2TB) daily at 99.9% accuracy and built Python reporting tools used by 10-12 engineers that cut manual effort by 60%. Collaborates with client stakeholders, engineering, and testing teams to translate requirements and user feedback into production-ready features and workflow improvements. Collaborates with client stakeholders, engineering, and testing teams to translate requirements and user feedback into production-ready features and workflow improvements.

SKILLS

Languages: Python, SQL, C++

Data & Analytics: pandas, NumPy, Tableau, ETL-style data pipelines, data validation & reconciliation, data modeling

Databases & Caching: PostgreSQL, SQL Server, Redis

Backend & APIs: REST APIs (FastAPI), distributed data processing, system design, performance tuning

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, Jenkins CI/CD, Git/Gerrit, Linux

Practices: Agile (Jira), code review, unit & integration testing, monitoring & alerting, root cause analysis, documentation

AI-Assisted Development: Claude and ChatGPT for code development, debugging, test generation, understanding unfamiliar code, and troubleshooting integrations

PROFESSIONAL EXPERIENCE

Sr. Software Engineer

Mar 2023 – Mar 2025

Tata Consultancy Services | Client: Nokia | Kolkata, India

- Built Python validation and reporting tools used by 10–12 engineers across development and testing teams, automating repetitive validation and reporting work, cutting manual effort by 60%.
- Supported and enhanced 100+ business-critical telecom data-processing workflows for high-volume CDR ingestion, validation, and downstream processing; optimized processing logic and SQL queries to improve performance by 35%.
- Built C++ and SQL data-processing applications handling 10M+ records (~2TB) daily across distributed nodes at 99.9% accuracy.
- Worked with client stakeholders, testing, and engineering teams to understand production issues and change requests, translate them into technical changes, and implement and validate fixes; turned client and QA feedback into application enhancements.
- Owned 10+ production releases end-to-end in an Agile/Jira environment, from technical design with client stakeholders and architects through Jenkins pipelines automating builds, tests, and artifacts across dev, QA, and production, with rollback procedures.
- Led 10–12 knowledge-transfer and technical walkthrough sessions for groups of 5–6 engineers on application workflows and troubleshooting; maintained support documentation and mentored junior engineers through code reviews.
- Operated high-availability services on AWS (EC2, S3) and Docker/Kubernetes with health checks, automated alerting, and secured service and database access; led root-cause analysis for 50+ production incidents.

Software Engineer

Mar 2021 – Feb 2023

Tata Consultancy Services | Chennai, India

- Developed C++ and SQL validation and reconciliation components with duplicate detection and replay-safe batch reprocessing.
- Built Git-based CI/CD build and release workflows that cut deployment time by 40% and enabled weekly releases.
- Shipped production fixes backed by unit and integration tests, cutting recurring issues by 25%.

PROJECTS

- A/B Test Ad Effectiveness Analysis** (SQL, Tableau): Compared ad variant performance in SQL; visualized results in Tableau.
- E-commerce Funnel Analysis** (SQL, Tableau): Measured stage-by-stage conversion and drop-off in SQL; visualized in Tableau.
- Customer Retention & Revenue Optimization Analysis** (Python, Tableau): Analyzed customer retention and revenue patterns.
- URL Shortener API** (Python, FastAPI, PostgreSQL, Redis, Docker): Built a versioned REST API with Redis TTL caching, cache invalidation, and rate limiting over PostgreSQL; containerized with Docker Compose and documented via OpenAPI. [GitHub](#)

EDUCATION

Trine University — M.S., Business Analytics (GPA 3.67/4.0)

May 2025 – Expected May 2027

Authorized for full-time work via CPT

Vellore Institute of Technology — M.Tech, Computer Science Engineering

Jun 2018 – Aug 2020

Christian College of Engineering & Technology — B.Tech, Computer Science

Jul 2013 – Aug 2017